

Hrithik Majumdar Shibu

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Summary

AI researcher and university lecturer with strong skills in Python, data analysis, NLP & LLMs, computer vision, statistical modeling, and applied machine learning. Experienced in undergraduate teaching, student mentoring, academic writing, research publication, and project supervision. Co-author of multiple peer-reviewed publications on fake news detection, online social harm, health AI, and agricultural data modeling. Focused on applying computational methods, responsible AI, and interdisciplinary research to real-world societal challenges.

Education

University of Essex, England – M.Sc. in Artificial Intelligence (Official Website) Sept 2025 – Ongoing

- **Coursework:** Deep Learning and Neural Networks, Machine Learning, Intelligent Systems and Robotics, Game Artificial Intelligence, Text Analytics, and Machine Learning for Finance.
- **Thesis:** Building an end-to-end Climate Well-Being NLP+ML pipeline using Transformers and LLMs with a hierarchical two-threshold gating classifier.

Shahjalal University of Science and Technology, Bangladesh – B.Sc. in Computer Science and Engineering Jan 2019 – Feb 2024

- CGPA: 3.61/4.00 (Official Website)
- **Coursework:** Discrete Mathematics, Structured Programming Language, Data Structure, Calculus, Algorithm Design & Analysis, Object-Oriented Programming Language, Numerical Analysis, Database Systems, Operating Systems, Networking, Computer Architecture, Data Science, Computer Security, Computer Graphics, Artificial Intelligence, and Machine Learning.
- **Thesis:** From Scarcity to Capability: Empowering Fake News Detection in Low-Resource Languages with LLMs.

Experience

Lecturer in CSE, R. P. Shaha University, Narayanganj, Bangladesh Sept 2024 – Ongoing

- Preparing the course outline, lesson plan, and tutorial, and conducting class lectures.
- Conducting and supervising students' projects and research studies.
- Supervising and guiding students' work and activities inside and outside the classroom.

Research Assistant, University of Essex, Colchester, United Kingdom April 2026 – Ongoing

- Performing high-level research on Multi-label Datasets and Classifiers for Low-Altitude Disaster Imagery under the Department of CSEE.
- Gaining hands-on experience in the domains of artificial intelligence and high-quality scientific writing and publishing articles in renowned journals.

Program and Research Coordinator, RPSU Research Club, Narayanganj, Bangladesh February 2025 – Ongoing

- Leading a whole research group of 10 students in various fields of Artificial Intelligence, Machine Learning, Computer Vision, and NLP.
- Published 7 high-quality articles and papers in renowned conferences associated with IEEE and ACL, and is continuously growing.

Data Annotator and Volunteer, Shahjalal University of Science and Technology, Sylhet, Bangladesh April 2019 – Nov 2022

- Annotated and analysed research projects named 'Emonoba: A dataset for analysing fine-grained emotions on noisy Bangla texts' and 'Bangla Word Stemming.'
- Organised LICT TECH FEST - 2019.
- Volunteered for a conference named ICBSLP-2019 and helped participants to present and publish their papers.

Publications

- SUST_Black Box at BLP-2023 Task 1: Detecting Communal Violence in Texts: An Exploration of MLM and Weighted Ensemble Techniques (BLP@EMNLP-2023)** Dec 2023
Hrithik Majumdar Shibu, Shrestha Datta, Zhalok Rahman, Shahrab Khan Sami, MD. Sumon Miah, Raisa Fairouz, Md. Adith Mollah
Key Focus: AI Safety · Social harm detection · Responsible NLP
- From Scarcity to Capability: Empowering Fake News Detection in Low-Resource Languages with LLMs (IndoNLP@COLING-2025)** Dec 2024
Hrithik Majumdar Shibu, Shrestha Datta, MD. Sumon Miah, Nasrullah Sami, Mahruba Sharmin Chowdhury, Md. Saiful Islam
Key Focus: Data-Centric AI · LLM Alignment · low-resource NLP
- Deep Learning Architectures for Real-Time Monkeypox Detection: A Comparative Study of CNNs, Transformers, and Siamese Networks (ICIT@IEEE-2025)** Dec 2025
Sakib Azmain Sahil, Badhan Rani Das, *Hrithik Majumdar Shibu*, and Aisha Hayder Chowdhury
Key Focus: Model Comparison · ViT · Model Safety
- Quantile-Based Crop Production Classification in Bangladesh Using Artificial Intelligence Techniques on Multisource Agro-Environmental Data (STI@IEEE-2025)** Dec 2025
Md. Siam, Navid Hasan Radiat, Mst. Afroza Akter, Dr. Kingkar Prosad Ghosh, and *Hrithik Majumdar Shibu*
Key Focus: Data curation · Structured prediction · Data-Centric AI
- Recognition of Fabric Construction Types with CNNs and Transformers: Analysis of Knit, Woven, and Non-Woven Fabrics (EICT@IEEE-2025)** Dec 2025
Shohag Babu, Meherun Nesa, Md. Selim, and *Hrithik Majumdar Shibu*
Key Focus: Benchmarking · Model Comparison · Dataset Curation
- Crop Production Prediction in Bangladesh on Quantile-Based Agricultural Data: A Comparative AI-Based Approach with Novel SAINT Model (ICIT@IEEE-2025)** Dec 2025
Md. Siam, Kingkar Prosad Ghosh and *Hrithik Majumdar Shibu*
Key Focus: Tabular DL · data-centric methods · Applied ML
- Visual Classification of Ten Region-Specific Flower Species in Bangladesh Using Hybrid CNN and Vision Transformers (ICIT@IEEE-2025)** Dec 2025
Sayma Islam, Ommey Habiba Sowme, Toukir Hossain, and *Hrithik Majumdar Shibu*
Key Focus: Model Efficiency · Data-Centric AI · Benchmarking

Languages

English: Reading:C1, Listening:B2, Writing:B2, Speaking:C1

Research Projects

- A Benchmark Bangla Fake News Dataset (IndoNLP @ COLING-2025)** github.com/Shibu4064/IndoNLP
- Created a robust dataset for Bangla fake news detection from over 60k articles (13K Fake and 47K Authentic news articles)
 - Improved fake news detection process implemented using Python, LLMs, BERTs, ML algorithms (SVM, LR), etc.
- Bangla Violence-Inciting Comment Detection using LLMs (BLP @ EMNLP-2023)** github.com/Shibu4064/EMNLP
- Built a Bangla NLP classifier for detecting violence-inciting YouTube comments using BanglaBERT, MLM fine-tuning, transfer learning, stacking, and weighted ensembles.
 - Achieved 71–72% macro-F1 in the BLP-2023, with the MLM-based system ranking 18th among participating teams.

Automatic Speech Recognition System using LoRA on Multilingual Datasets github.com/Shibu4064/ASR
(IEEE ICCSC-2026)

- Fine-tuned Whisper-Small, XLS-R 300M, Conformer-CTC, and QuartzNet 5×5 on LibriSpeech and Mozilla Common Voice in 10 different languages.
- Applied LoRA/PEFT, reducing trainable parameters from 244M to 4.7M with minimal WER degradation for an end-to-end system deployment.

Social Harm & Multi-Agent NLP Research on Social Media Contents (IEEE PECCII-2026) github.com/Shibu4064/Violence

- Led the construction of a multi-source Bangla violence and hate speech dataset, addressing data curation challenges at scale, which is foundational to training safer, aligned NLP systems.
- Applied ensemble NLP methods and LLM prompting strategies to multi-label classification of harmful content, touching on multi-agent LLM dynamics and AI safety.

Technical Skills

AI/ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face, NumPy, Pandas
ML & DL	Model training and evaluation, feature extraction, classification, optimisation
NLP & LLMs	BERTs, LLMs (Bloom, Phi3, LLaMa), transformers, NER, sentiment analysis, LLM prompting, language model
Computer Vision	OpenCV, YOLOv5, image preprocessing, object detection, visual recognition
Responsible AI	Dataset curation and auditing, fairness-aware evaluation, explainability techniques, data attribution, bias-aware assessment
Programming	Python, C, C++, Java, SQL, PHP
Software & Web Development	Django, Laravel, Node.js, Express.js, Next.js, HTML, CSS, JavaScript
Tools & Writing	Git, Jupyter, Microsoft Office, Google Docs/Sheets, PowerPoint, scientific writing, technical documentation

Extracurricular Activities

Public Relations Coordinator: Organized HULT PRIZE 2022-23 at SUST and participated in the Bangladesh Math Olympiad 2018.

Vice President: Carried out responsibilities as a Vice President of the Greater Faridpur Association of SUST.

Editor: Honored to carry out my responsibilities as an editor of a magazine for Ramakrishna Mission, Dhaka.

Operational Executive: Carried out my duties as an operational executive at SUST ACM Student Branch Chapter.

Certifications & Awards

Supervised Machine Learning: Classification and Regression github.com/Shibu4064/Certificates

Advance Learning Algorithms by Coursera

Exploratory Data Analysis for Machine Learning by Coursera

Introduction to Research for Essay Writing by Coursera

Business Analysis and Process Management by Coursera

Best Employee Ranking Award-2025 by R. P. Shaha University

References

Dr Sangeet Saha, Lecturer, Dept. of CSEE, University of Essex sangeet.saha@essex.ac.uk
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Dr Delaram Jarchi, Lecturer, Dept. of CSEE, University of Essex delaram.jarchi@essex.ac.uk
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